

1.4.27

EE25BTECH11011 - Navya Banavathu

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Question

If $\mathbf{a}$ and $\mathbf{b}$ are the position vectors of A and B respectively, find the position vector of a point C in BA produced such that $BC = 1.5 BA$

Solution

Given the position vectors of **A** and **B** are **a** and **b**

Let the position vector of point **C** be **c**

The vector BC is given by:

$$\mathbf{BC} = 1.5\mathbf{BA} \quad (1)$$

$$(\mathbf{c} - \mathbf{a}) : (\mathbf{b} - \mathbf{c}) = 0.5 : 1.5 = 1 : 3 \quad (2)$$

Therefore, *C* divides *AB* externally in the ratio 1 : 3

Writing the ratio as *k* : 1 we have

$$k = 3 \quad (3)$$

Section Formula(external division)

By the section formula (external division), the position vector of C is

$$\mathbf{c} = \frac{k\mathbf{b} - \mathbf{a}}{k - 1}, \quad (k = 3). \quad (4)$$

$$\mathbf{c} = \frac{3\mathbf{b} - \mathbf{a}}{2}. \quad (5)$$

$$\therefore \mathbf{C} = \frac{3}{2}\mathbf{b} - \frac{1}{2}\mathbf{a}. \quad (6)$$

The above equation can be written as:

$$\mathbf{c} = (\mathbf{a} \quad \mathbf{b}) \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix} \quad (7)$$

C Program

```
#include <stdio.h>

int main() {
    float ax, ay, bx, by;
    printf("Enter coordinates of A (ax ay): ");
    scanf("%f %f", &ax, &ay);
    printf("Enter coordinates of B (bx by): ");
    scanf("%f %f", &bx, &by);
    int k = 3;
    float cx = (k*bx - ax) / (k - 1);
    float cy = (k*by - ay) / (k - 1);
    printf("Position vector of C is: (%.2f, %.2f)\n", cx, cy);
    return 0;
}
```

Python Code: Import and Define points

Part 1

```
import matplotlib.pyplot as plt

# --- Data ---
# Position vectors (points A and B)
A = (2, 3) # Example coordinates of A
B = (5, -1) # Example coordinates of B

# Ratio for external division (C divides AB externally in 3:1)
k = 3

# Section formula for external division:
#  $C = \frac{k \cdot B - A}{k - 1}$ 
Cx = (k*B[0] - A[0]) / (k - 1)
Cy = (k*B[1] - A[1]) / (k - 1)
C = (Cx, Cy)

print(f"Position vector of C is: {C}")
```

Python Code: Plotting

Part 2

```
# --- Plotting ---
points = [A, B, C]
x_coords, y_coords = zip(*points)

plt.figure(figsize=(7, 6))

# Plot points
plt.scatter(x_coords, y_coords, color='red', s=80, zorder=5)

# Connect A-B (main line)
plt.plot([A[0], B[0]], [A[1], B[1]], color='blue', label='Line AB
        ')

# Show extended line through A-B-C
plt.plot(x_coords, y_coords, '--g', label='Line with C')
```

Python Code: Finishing Plot

Part 3

Labels

```
labels = ['A'+str(A), 'B'+str(B), 'C'+str(C)]
for (x, y), label in zip(points, labels):
    plt.text(x + 0.2, y + 0.2, label, fontsize=10)
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.gca().set_aspect('equal', adjustable='box')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('External Division (C divides AB in ratio 3:1)')
plt.legend()
plt.grid(True)
import os
# Your plotting code above...
plt.savefig("figs/fig1.png", dpi=300, bbox_inches="tight")
plt.show()
```


Graphical Representation

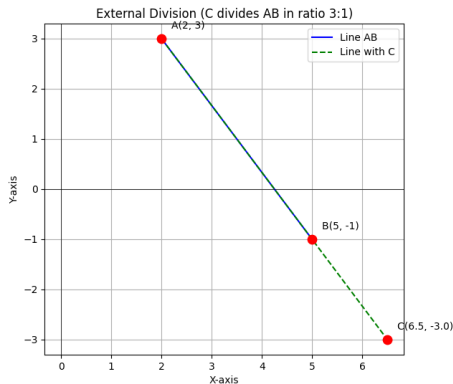


Figure: Point C dividing AB externally in ratio $3 : 1$